



Telecommunication Churn

Classification Project

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StreamLit link :

<https://excelr-final-project-telecom-churn.streamlit.app>

GitHub link :

https://github.com/Sourabh-Biradar/ExcelR_Final_Project/tree/main

Overview

High **customer churn** is endemic to the telecommunications sector, with typical annual rates surpassing **10%**. This critical business challenge necessitates **robust client retention strategies**. Our work addresses this through a **binary classification project**, focused on predicting the key outcome of whether a customer will **churn or remain loyal**.

Goals

1. The primary goal is to build, evaluate, and optimize a **binary classification** model that accurately predicts future customer behavior—specifically, distinguishing between **loyal customers** and those who will **churn**.
2. To construct a highly accurate **predictive classification model** (Churn/Loyal) that enables the company to proactively intervene with targeted strategies, thereby reducing the annual **churn rate** (currently >10%) and improving overall business stability.

Data Source

- Dataset : 'P585 Churn.xlsx'
- Target variable : '**churn**' (no/yes)

Methodology

Data Exploration and Preprocessing (EDA)

- Initial Data State : The dataset contained **5000 records and 21 features**.
- Target Imbalance : '**no**' [4293 (85.86%)] & '**yes**' [707 (14.14%)]
- Cleaning Steps :
 - The dataset had **no missing or duplicate** records.
 - There were **2 miss categorised** numerical features '*day.charge*' & '*eve.mins*', handled using Pandas in-built method '**to_numeric**'.
 - Every feature had outliers , used '**Manual Clipping**' with limits (lower&upper).
 - '*voice.messages*', '*intl.calls*' features were heavily **Right skewed** so applied 'log' transformation on them.
 - Further '*voice.messages*' was scaled using RobustScaler as it was still right skewed

- Ran **chi-square test of independence** on categorical features to determine their association with target variable
- Feature Engineering :
 - Created a **new feature** `['total.mins']` merging all 'mins' features.
 - Likewise created few **new ratio features** & dropped original base features.
 - `['eve.charge_per_min']` had some '**inf**' values imputed with '0'.
 - Dropped **highly correlated** features.
- Data Visualization :
 - **Box-plots** : for outliers analysis (pre & post Clipping).
 - **Histograms** : for distribution & skew analysis.
 - **HeatMaps** : to check *Multicollinearity*.
 - **Pair Plots** : for *Univariate (The Diagonal Plots) & Bivariate Analysis (Non-Diagonal plots)*.
 - **Count plots** : for distribution of categorical columns.
- Train-Test split :
 - **test_size=0.2** : Test set 20% and Train set 80%.
 - '**stratify=y**' : to ensure that the *target class distribution* is *identical* in both training and testing datasets.
 - Used **SMOTE** technique to handle imbalance in target class (*Over-sampled training set*) inside **pipeline**.
- Preprocessing :
 - **Label Encoded** target variable '*churn*'.
 - **Frequency Encoded** '*state*' feature as it has 51 classes , too much to OneHotEncode as it would lead to overfitting & False weightage if Ordinal Encoded.
 - **Ordinal encoded** remaining categorical features.
 - *Standardization* of numerical features using **Standard Scaler** (Z-score normalization).
 - **Robust Scaled** '*voice.messages*'

Model training and Hyperparameters tuning

Used GridSearchCV & RandomizedSearchCV with cv = 5 (cross validation k-folds)

Pipeline >> SMOTE + models

- Best params for **Logistic Regression** model
 - $C = 0.1$,
 - $penalty = 'l1'$,
 - $solver = 'saga'$
- Best params for **Decision Tree Classifier** model
 - $criterion = 'entropy'$,
 - $max_depth = 5$,
 - $max_features = None$,
 - $min_samples_leaf=15$,
 - $min_samples_split=10$
- Best params for **K-Nearest Neighbors Classifier** model
 - $n_neighbors = 31$,
 - $weights = 'distance'$,
 - $algorithm = 'auto'$,
 - $p = 1$,
 - $metric='minkowski'$ ($minkowski_distance(l_p)$: $manhattan_distance(l1)$ and $euclidean_distance(l2)$)
- Best params for **Support Vector Classifier** model
 - $C = 1$,
 - $gamma = 'auto'$,
 - $class_weight= 'balanced'$
 - $kernel = 'rbf'$
- Best params for **Random Forest Classifier** model
 - $n_estimators=300$,
 - $max_depth=20$,
 - $min_samples_leaf=2$,
 - $min_samples_split=5$,
 - $max_features='sqrt'$

- Best params for **XGBoost Classifier** model
 - *n_estimators=200,*
 - *max_depth=15,*
 - *min_child_weight=1,*
 - *gamma=0.0,*
 - *colsample_bytree=1.0,*
 - *subsample=0.9*
- Best params for **LightGBM Classifier** model
 - *n_estimators=200,*
 - *max_depth=7,*
 - *learning_rate=0.1,*
 - *num_leaves = 31,*
 - *subsample=1.0*

Model evaluation on training set

Models	ROU-AUC score
Logistic Regression	0.80
Decision Tree Classifier	0.89
K-Neighbors Classifier (KNN)	0.82
Support Vector Classifier (SVC)	0.87
Random Forest Classifier (Bagging)	0.919
Extreme Gradient Boosting Classifier (XGB)	0.924
Light Gradient Boosting Classifier (LightGB)	0.921

NOTE : Used '**roc_auc**' as the scoring metric in GridSearchCV & RandomizedSearchCV to find the best combination of hyperparameters

Model evaluation on test set

Model Performance Comparison

Models	ROU-AUC score	Precision	Recall	F1 score
Logistic Regression	0.795	0.32	0.72	0.44
Decision Tree Classifier	0.918	0.73	0.69	0.71
K-Neighbors Classifier (KNN)	0.804	0.34	0.72	0.47
Support Vector Classifier (SVC)	0.89	0.51	0.81	0.63
Random Forest Classifier (Bagging)	0.912	0.87	0.77	0.82
Extreme Gradient Boosting Classifier (XGB)	0.915	0.89	0.80	0.84
Light Gradient Boosting Classifier (LightGB)	0.928	0.88	0.74	0.80

BEST MODEL : *Extreme Gradient Boosting Classifier (XGB)*

NOTE : Precision , Recall & F1 score are for 'churn' (Class : 1)

Evaluation Metrics

- **Area Under the ROC Curve (AUC):** A measure of the model's ability to distinguish between the two classes across various thresholds.
- **Recall (Sensitivity):** The ability of the model to correctly identify *all* customers who will churn (minimizing False Negatives, the most costly error).
- **Precision:** The accuracy of the model's positive predictions (how many predicted churners *actually* churned).
- **F1-Score:** The harmonic mean of Precision and Recall.

Conclusion on Performance :

The *Extreme Gradient Boosting Classifier (XGB)* classifier demonstrated superior performance, achieving the best balance between identifying actual churners (Recall) and maintaining reliable positive predictions (Precision). Its Recall of **0.80** means that the model successfully captured **80%** of all customers who ultimately left the company.

Key Feature Importance

The model provided significant insight into the main drivers of customer churn. The top 4 features influencing the prediction were:

1. **total.mins** : High Usage = High Churn Risk
2. **customer.calls** : High calls = High Churn Risk
3. **intl.plan** : active international plan = High Churn Risk
4. **voice.plan** : active voice plan = Lesser Churn Risk

Conclusion and Recommendations

Project Conclusion

The classification project successfully addressed the challenge of telecommunications customer churn. The optimized **XGBClassifier** model provides the company with a powerful tool to predict customer attrition with an effective Recall of **80%**. This level of predictive capability enables the transition from reactive to proactive client management.

Business Recommendations

Based on the model results and feature analysis, the following actions are recommended for the telecom company:

1. **Targeted Retention Campaigns :**

Deploy personalized retention offers (e.g., loyalty discounts, custom plans) for customers identified with *high total usage minutes and frequent customer service calls*, as these segments show a higher likelihood of churn. Tailored engagement and proactive support may reduce dissatisfaction among heavy users.

2. **Contract Structure Review :**

Focus retention efforts on customers with Month-to-Month contracts by offering incentives to switch to long-term contracts (e.g., annual or biannual plans). Longer commitments help stabilize the customer base and reduce churn volatility.

3. **Plan Optimization Strategy :**

Customers with an *active international plan* exhibit *higher churn risk* — likely due to high costs or dissatisfaction with international call rates. Reassess pricing or provide competitive international bundles to retain this segment.

Conversely, customers with an *active voice plan* are *less likely to churn*. Promote or cross-sell voice plans to at-risk customers as part of a retention bundle.

4. **Service Enhancement :**

Enhance the quality and reliability of customer support and technical services, especially for users making *frequent customer calls*. Quick resolution and personalized care can directly reduce churn among this group.

5. **Early Intervention Program :**

Implement a “first 90 days” loyalty program for new or low-tenure customers. Proactive communication, onboarding support, and early engagement can help prevent churn before dissatisfaction develops.